

Youssef Naciri Zerta

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EMBL-EBI, Hinxton — Saez Lab, | Seeking a full-time research thesis, February–August 2027

RESEARCH FOCUS

I work on **causal representation learning for gene regulation**, and for systems biology more broadly. The question I care about is what a representation of a biological system has to satisfy in order to predict behaviour *under intervention*, rather than only describe state. I pursue it from three directions at once: building the infrastructure that makes causal inference over biological structure from prior knowledge and omics data; testing whether existing representational claims survive interventional ground truth, which so far they have not; and developing an account of regulatory structure that does not rest on the dimension-counting and discrete-module formalisms the field currently assumes. I have worked on each of these in a different setting: EMBL-EBI (Open Targets), Helmholtz Munich, and my own experimental work on microbial regulation (UEMF, alma mater); and the aim is to bring them into one programme.

RESEARCH

SysBioVerse — Lead Developer

2026–Present

Open Targets — EMBL-EBI, Saez Lab – *ecosystem for causal network inference*

- A unified ecosystem for methods that infer causal and mechanistic biological structure from omics data and prior knowledge: the intended equivalent of scverse for causal network inference.
- Designing a common interface across heterogeneous causal-inference workflows so that regulatory, signalling, metabolic, cell-cell and multi-omic methods can be compared, benchmarked and composed within one system, rather than reimplemented per paper against incompatible data structures.
- The design problem is a representational one: what abstraction is shared by methods that disagree about what a causal biological structure is, and what has to be exposed for their outputs to be comparable at all.

AnnNet — Lead Developer

2025–Present

EMBL-EBI, Saez Lab – *annotated biological network structure; co-first-author preprint in preparation (Bioinformatics)*

- [AnnNet](#) is a Python data structure for annotated biological networks supporting directed, signed, multilayer, hypergraph and condition-specific topology under a single interface, the classical graph formalism is just a special case rather than the frame.
- Designed the sparse incidence-based core, the typed annotation model, native storage, and interoperability with the major graph libraries.
- Motivated by a concrete limitation: biological structure is routinely multilayer and context-dependent, and forcing it into a single directed graph discards exactly the conditioning that makes a network claim causal.

SampleCLR — Patient-Level Representations from Single-Cell Transcriptomics

2025

Helmholtz Munich, Lücken Lab – *contrastive and attention-based sample representation* [\[preprint\]](#)

- Contributed interpretability and transfer-learning analyses, and to the architecture design, for a model producing patient-level representations from single-cell data with state-of-the-art disease-stratification performance.
- Evaluated whether the learned representations transferred across biological systems rather than only fitting the training benchmark — the question of whether a representation generalises beyond its benchmark is the one my later work formalises.

Interventional Testing of Cross-Modal Representational Convergence

2026

Sole author — in preparation, ICLR 2027 – *title withheld for double-blind review*

- A widely-reported measure of cross-modal representational convergence is read against a baseline that is itself capacity-dependent. Derived the corrected null in closed form, and showed that once it is subtracted the reported scaling trend does not survive.
- Decided the question on a biological testbed rather than a vision–language one: a coding variant is two exact views of a single object, related by the genetic code and labelled by saturation genome editing, so every term of the null is measured rather than assumed. Using a causal biological assay to settle a question the machine-learning literature could not is the method I want to generalise.

A Causal Signal Without Causal Features

2026

Sole author — under review, MLCB 2026 (Machine Learning in Computational Biology) – *identifiability of sparse-autoencoder features in genomic foundation models* [\[preprint\]](#)

- Tested whether sparse-autoencoder features of genomic foundation models correspond to reproducible regulatory structure, using saturation-mutagenesis MPRA (massively parallel reporter assay) as causal ground truth.

- A causal signal is present in aggregate, but the decomposition carrying it is not identifiable — across retraining, across autoencoder families, or against a supervised control. This is a constraint on any latent account of gene regulation rather than a verdict on one method.

Multiplex Bacterial Interaction Networks

2024–Present

Université Euromed de Fès – *experimental systems biology; first author, in preparation (ISME J)*

- Designed and executed a trait-resolved interaction study across 60 soil bacterial strains, generating 12 directed interaction layers from pairwise assays — an interaction tensor rather than a single network.
- Built a multilayer-network analysis of that tensor to characterise microbial regulatory strategies across ecological contexts, showing that interaction sign and strength are context-dependent rather than intrinsic to a strain pair.
- This is where the limitation that motivates AnnNet came from: a single directed graph cannot represent a system whose edges change meaning with condition.

Epigenomic Biomarker Combination Detection

2024–2025

MD Anderson Cancer Center / KAUST – *analysis pipeline*

- Built a pipeline for detecting combinations of biomarkers from epigenomic data.

PUBLICATIONS & PREPRINTS

- **Y. Naciri Zerta.** *A Causal Signal Without Causal Features: identifiability of sparse-autoencoder features in genomic foundation models.* Sole author, under review, MLCB 2026. [\[preprint\]](#)
- *Great patients embed alike: contrastive learning for sample representation from single-cell data.* Middle author. [\[preprint\]](#)
- *AnnNet: a unified annotated-graph structure for biological networks.* Co-first author, in preparation (Bioinformatics).
- *Patterns of pairwise interactions between Streptomyces spp.* First author, in preparation (ISME J).

EDUCATION

State Engineer (Diplôme d'Ingénieur d'État) – Biotechnology

Expected June 2027

Université Euromed de Fès (UEMF), Morocco – *five-year engineering degree; Master's-equivalent, 300 ECTS*

Relevant coursework: Systems Biology, Synthetic Biology, Recombinant DNA Technology, Advanced Microbiology, Statistical Genetics, Optimization, Probability & Statistics.

METHODS

Causal & representation learning: causal representation learning, identifiability theory, interventional and counterfactual inference, sparse and overcomplete latent-variable models, mechanistic interpretability, foundation models.

Statistical modelling: high-dimensional and sparse inference, spectral methods, stochastic dynamics and linear-response theory.

Evaluation design: interventional ground truth, group-structured cross-validation, leakage diagnosis, floor and ceiling calibration, planted nulls, identifiability diagnostics.

Genomics & regulation: gene regulatory networks, perturbational genomics, single-cell transcriptomics, statistical genetics.

Systems biology: mechanistic modelling, dynamical systems, multilayer and hypergraph formalisms, prior-knowledge integration.

Engineering: Python/PyTorch, R, C/C++, scientific library design, open-source maintenance, GPU training, cloud computing, Docker.

Wet lab: molecular cloning, genome engineering, microbial culture, analytic chemistry.

Languages: English & French (fluent), Arabic & Tamazight (native), Spanish (basic).